

Task Overview

In this task, you'll be introduced to the Splunk platform. Then, you'll use Splunk Cloud to upload data, perform basic searches on the data, and answer a series of questions. Before you start using Splunk, you will need to do the following:

- Create a Splunk account
- Activate a free trial of Splunk Cloud
- Upload data into Splunk Cloud

So far, you've learned that SIEM tools, such as Splunk, are an important part of a security analyst's toolbox because they provide a platform for storing, analyzing, and reporting on data from different sources.

Creating effective searches is an important skill because it enables you to quickly and accurately find the information you are looking for within a large amount of data. Quick and accurate searching is especially useful during incident response, because you might need to swiftly identify and address a security incident. Effective search techniques also help you efficiently identify patterns, trends, and anomalies within data.

Scenario

Review the following scenario. Then complete the step-by-step instructions.

You are a security analyst working at the e-commerce store Buttercup Games. You've been tasked with identifying whether there are any possible security issues with the mail server. To do so, you must explore any failed SSH logins for the root account.

Step-By-Step Instructions

Follow the instructions and answer the following questions to complete the task.

Step 1: Access supporting materials

The following supporting materials will help you complete this task. The data contains log and event information from Buttercup Games' mail servers and

web accounts. This includes information like access and authentication logs, email logs, and more.

Step 2: Create a Splunk account

To use Splunk Cloud, you must create an account. Follow these steps to create an account.

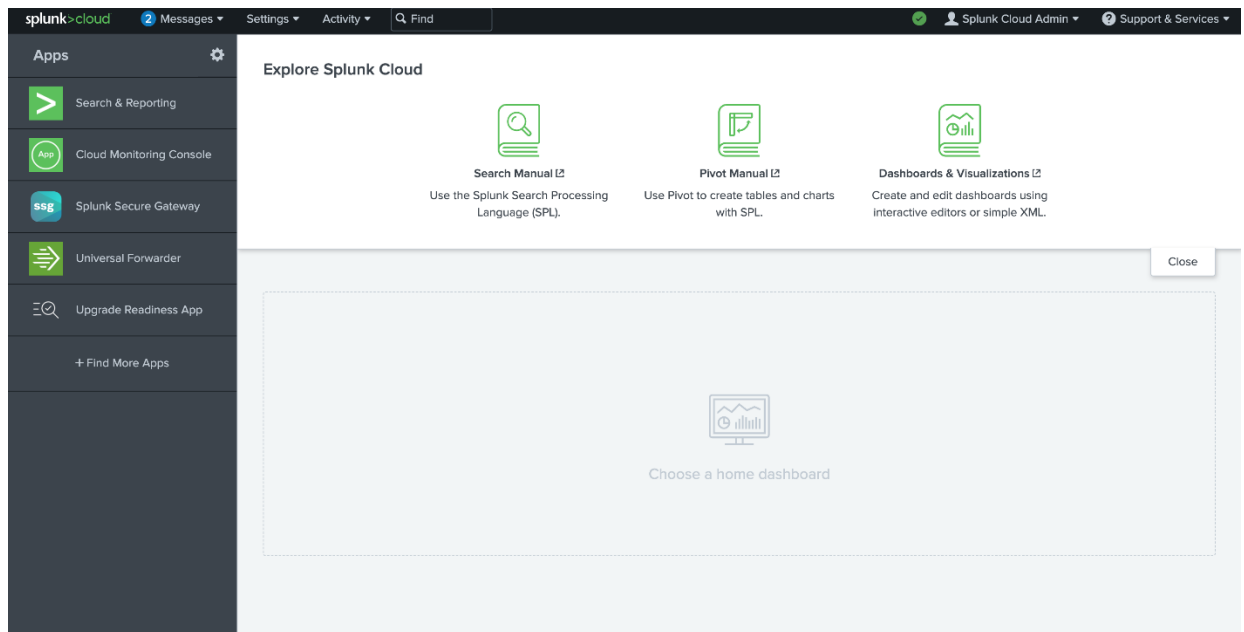
1. Visit [Splunk Cloud Platform Trial](#) and complete the sign-up form to create a free account.
2. After completing the required information on the sign-up form, click **Create your account**.
3. Check your email inbox for a verification email from Splunk. Follow the instructions in the email to verify your email address and get started.

***Note:** Check your spam folder if you didn't receive the verification email.*

Step 3: Sign up for a free Splunk Cloud trial

After you've created your Splunk account, you'll need to sign up for a free Splunk Cloud trial.

1. Visit the link [Splunk Cloud Platform Trial](#) and sign in to your Splunk account.
2. Click **Start Trial**.
3. Check your email inbox for an email from Splunk. The email will contain the following information:
 - Login information for the Splunk Cloud instance, including a username and a temporary password
 - A link to the Splunk Cloud instance
4. Click the link in the email from Splunk to access the Splunk Cloud instance.
5. Log in to the instance using the credentials contained in the email from Splunk. You will be prompted to choose a new password to replace the temporary password.
6. Once you've logged in, you'll automatically be redirected to the Splunk Home dashboard. From here, you'll begin the task.



***Note:** You can activate one Splunk Cloud trial instance at a time, and you can use a maximum of three trials per Splunk account. The Splunk Cloud free trial expires after 14 days, so you may want to complete this task before the free trial expires.*

Step 4: Upload data into Splunk

To operate effectively, it's essential that SIEM tools ingest and index data. SIEM tools collect and process data so that it becomes searchable events that can be queried, viewed, and analyzed.

So far, you've created a Splunk account and activated and accessed the Splunk Cloud free trial, but your Splunk Cloud instance does not contain any data. Next, you'll need to upload data into Splunk to start querying. Complete the following steps to upload data into Splunk:

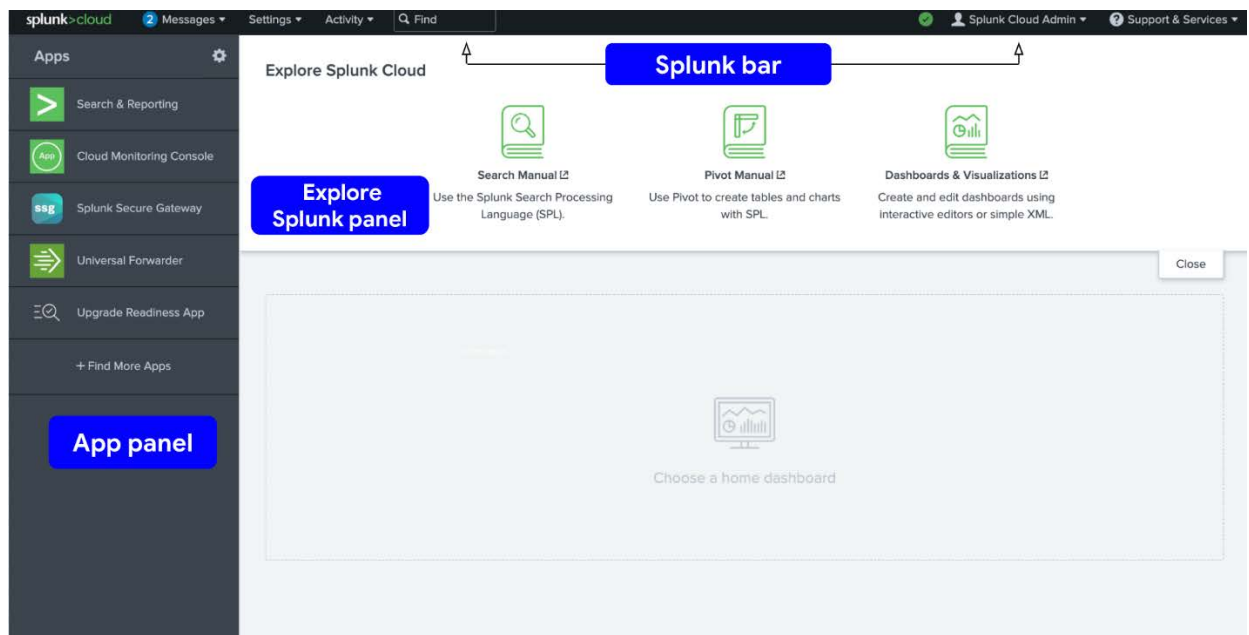
1. If you haven't already, download the data file from Step 1. Do not uncompress the file.
2. Navigate to Splunk Home from your Splunk Cloud free trial instance. You might need to log in again using your credentials from Step 3.
3. On the Splunk bar, click **Settings**. Then click the **Add Data** icon.
4. Click **Upload**.
5. Click the **Select File** button.
6. Upload the **tutorialdata.zip** file, and click **Open**.
7. Click the **Next** button to continue to **Input Settings**.
8. By the **Host** section, select **Segment in path** and enter **1** as the segment number.
9. Click the **Review** button and review the details of the upload before you submit. The details should be as follows: Input Type: Uploaded File File

Name: tutorialdata.zip Source Type: Automatic Host: Source path segment number: 1 Index: Default

10. Click **Submit**. Once Splunk has ingested the data, you will receive confirmation that the file was successfully uploaded.

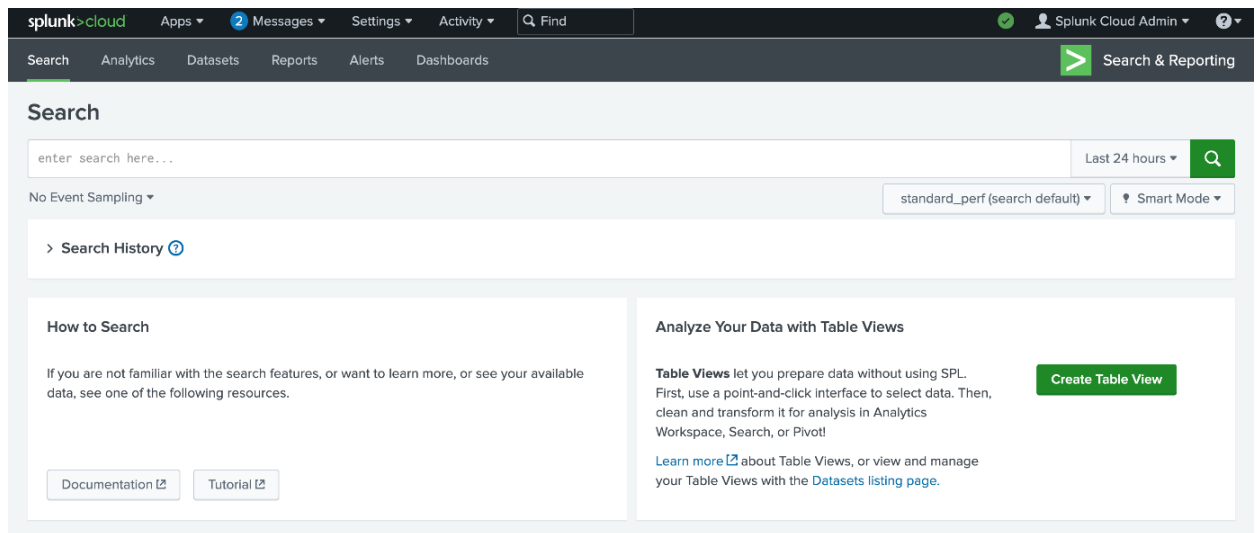
Step 5: Perform a basic search

Take a moment to examine the Splunk Cloud interface by locating the app panel, the Explore Splunk panel, and the Splunk bar.



After you've uploaded the data into Splunk, perform your first query to confirm that the data has been ingested, indexed, and is searchable. Follow these steps to perform a query:

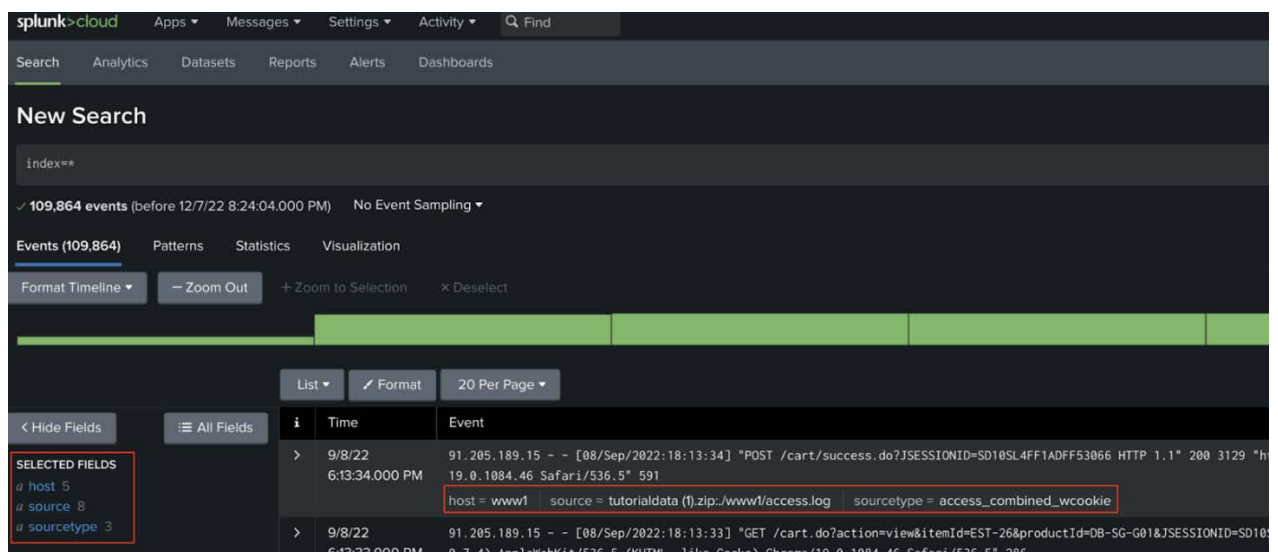
1. Navigate to Splunk Home. (To return to Splunk Home, click the Splunk Cloud logo on the Splunk Cloud page.)
2. Click **Search & Reporting**. You may close any pop ups that appear.
3. In the search bar, enter your search query: **index=main** This search term specifies the index. An **index** is a repository for data. Here, the index is a single dataset containing events from an index named main.
4. Select **All Time** from the time range dropdown to search for all the events across all time.
5. Click the search button. Note that the search button is represented by the magnifying glass icon. Your search should retrieve thousands of events.



***Tip:** It's a best practice to use short time ranges in your searches because a shorter time range returns results faster and uses fewer resources. Adjust the time using the time range dropdown or by using [time modifiers](#) in your search.*

Step 6: Evaluate the fields

When Splunk indexes data, it attaches fields to each event. These fields become part of the searchable index event data. This helps security analysts easily search for and find the specific data they need. Now that you've run your first query, examine the search results and the fields.



For each event the fields are **host**, **source**, and **sourcetype**. Under **SELECTED FIELDS**, examine the same fields.

Examine the field values by clicking on the field under **SELECTED FIELDS**. You should observe the following:

- **host**: The host field specifies the name of the network host from which the event originated. In this search there are five hosts:
 - **mailsv** - Buttercup Games' mail server. Examine events generated from this host.
 - **www1** - This is one of Buttercup Games' web applications.
 - **www2** - This is one of Buttercup Games' web applications.
 - **www3** - This is one of Buttercup Games' web applications.
 - **vendor_sales** - Information about Buttercup Games' retail sales.
- **source**: The source field indicates the file name from which the event originates. You should identify eight sources. Notice **/mailsv/secure.log**, which is a log file that contains information related to authentication and authorization attempts on the mail server.
- **sourcetype**: The sourcetype determines how data is formatted. You should observe three sourcetypes. Examine **secure-2**.

Step 7: Narrow your search

Because you've been tasked with exploring any failed SSH logins for the root account on the mail server, you'll need to narrow the search results for events from the mail server.

Under **SELECTED FIELDS**, click **host** and click **mailsv**.

Notice that a new term has been added to the search bar: **index=main host=mailsv**. The search results have narrowed to over 9000 events that are generated by the mail server.

How to pass Splunk quiz. Brief description.
Solve every Use Case in Splunk Cloud, get results and put them as an answer in proper quiz question. Could be done in parallel way or just write down results of Use Case Searches.

Use Case1(example): Search for a failed login for root

Now that you've narrowed your search results to events generated by the mail server, continue to narrow the search to locate any failed SSH logins for the root account.

1. Clear the search bar.
2. Enter **index=main host=mailsv fail* root** into the search bar. This search expands on the search from the previous task and searches for the keyword **fail***. The wildcard tells Splunk to expand the search term

to find other terms that contain the word *fail* such as *failure*, *failed*, etc. Lastly, the keyword **root** searches for any event that contains the term root.

3. Set up Date and Time Range

(ALL SEARCHES SHOULD USE THE SAME Date and Time Range)

The screenshot shows a configuration window for 'Date time range'. It has a search icon in the top right. The main area has a list of options: 'Presets', 'Relative', 'Date Range', 'Date & Time Range' (which is expanded and highlighted with a blue border), and 'Advanced'. Inside the 'Date & Time Range' section, there is a 'Between' dropdown, two date-time input fields, and an 'Apply' button. The first input field contains '03/01/2023' and '12:00:00.000', and the second contains '03/02/2023' and '12:00:00.000'. Both time fields are underlined in red. Below the time fields, the format 'HH:MM:SS.SSS' is displayed.

4. Click **search**.

Found number of events is a quiz answer for the proper question. (41) [visible for students]

Use Case2: Search for a failed login for root on host “www1” (**) at the same Date and Time Range. [hidden from students, part of quiz]

Use Case3: Search for a failed login for root on host “www2” (**) at the same Date and Time Range. [hidden from students, part of quiz]

Use Case4 (example): Search for a HTTP responses status “Client side error”

sourcetype=access_* status=40*

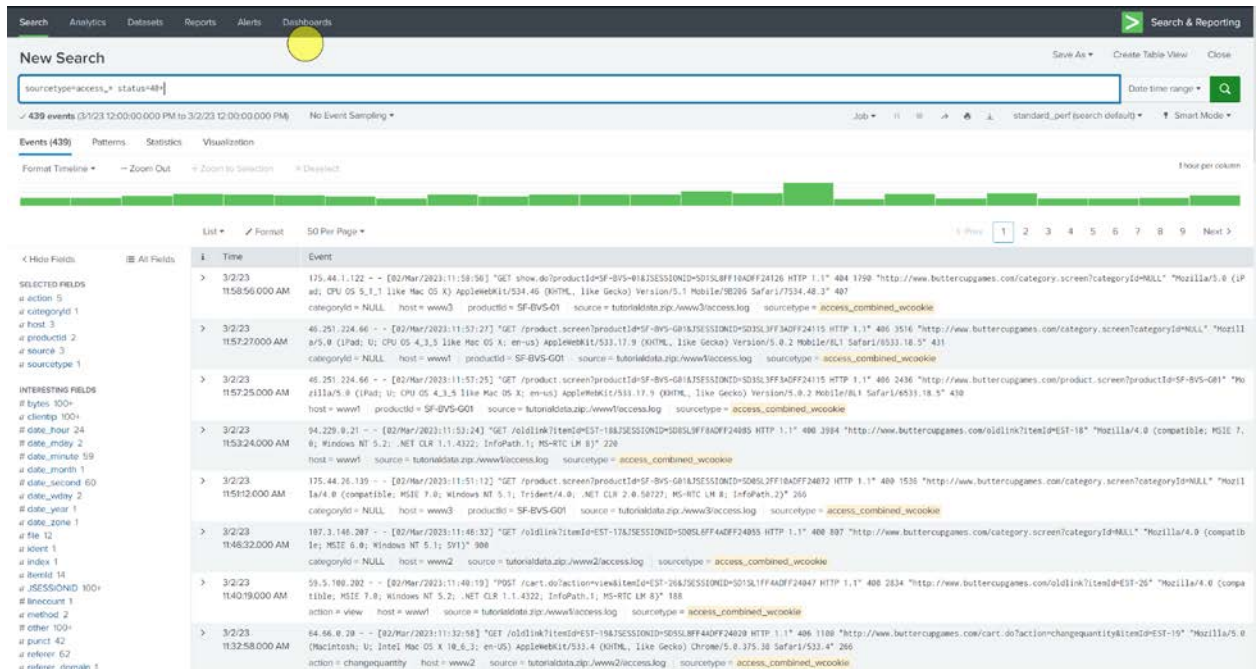
Check number of Events(439) [visible for students]

Use Case5 : Search for a HTTP responses status “Client side error” from “www1” host.

Check number of Events(***) [hidden from students, part of quiz]

Use Case6 : Search for a HTTP responses status “Client side error” from “www3” host.

Check number of Events(***) [hidden from students, part of quiz]



Use Case7 (example): Search for a HTTP responses status “Server-side error”

`sourcetype=access_* status=50*`

Check number of Events(306) [visible for students]

Use Case8 : Search for a HTTP responses status “Server-side error” from “www2” host

Check number of Events(**) [hidden from students, part of quiz]

Use Case9 : Search for a HTTP responses status “Server-side error” from “www3” host

Check number of Events(***) [hidden from students, part of quiz]

Use Case10 (example) : Search for a SSH accepted login attempts `sourcetype="secure-2" accept*`

Check number of Events(229) [visible for students]

Use Case11 : Search for a SSH failed login attempts
Check number of Events(****) [hidden from students, part of quiz]

Use Case12

Search for a SSH failed login attempts for host “www2”
Check number of Events(****) [hidden from students, part of quiz]

Use Case13

Search for a SSH accepted login attempts for host “www3”
Check number of Events(**) [hidden from students, part of quiz]

Use Case14

Search for a SSH accepted login attempts for host “mailsv”
Check number of Events(**) [hidden from students, part of quiz]

Step 8: Go to the module SIEM and take a quiz about achieved results

Congrats!